

SINKGRAB · CALCULATION

SinkGrab sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	First issue for TRL 3 on the constructable design (SKG-DDR-002); grab capacity and closing force, overload limits, crank force and speed, drum and rope, brake, tripod and drawbar, scraper, sinking and tilt, cycle time, masses and cost

SinkGrab sizing calculations

On paper SinkGrab does what it was drawn for, inside the HatchSide tripod's rating, but not in the 3 min its cycle-time requirement asks for. The grab lifts 21 L of saturated sand a bite at an assumed 75 % fill, two people hoist the full 93 kg grab with 52 N each on the cranks, a shear link and a crank shear pin keep the line below the tripod's 150 kg rating and 225 kg proof load whatever the crew does, and the heaviest piece is 24.6 kg. The parts cost USD 2,506 against a value-engineering target of USD 4,000. The miss is R4: a cycle at 10 m takes about 3.7 min, because two people give about 100 W and hoisting the full grab takes 111 s. Two requirements are at risk: R3, because the lips close with only about 0.7 of the line pull and the fill in denser sand is uncertain, and R1, because the rings keep sinking to 3 m only while skin friction stays below about 1.7 kPa. Seven are met on paper or by design and R6 can only be shown in a trial. Every number here is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [C2], is the line of that script's output that carries it.

Safety: SinkGrab is lifting equipment over an open well shaft. These are first-principles estimates for a paper proof of concept; they do not show that any part is safe. Every lifting part must be proof-loaded before use (SKG-BLD-001, section 6), nobody stands under the grab or beside a loaded line, and nobody enters the well. See SKG-PRC-001, Safety.

■ Scope and method

The note checks every requirement in SKG-REQ-001 v0.3 against the design in SKG-PRC-001 v0.3 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS`, `derived()` and `masses()`, so the grab, drum, scraper and frame used here are the ones in the STEP files and drawings SKG-DWG-001 and 002. It also reads `bom/bom.csv` and `budget_usd` in `project.yaml`, and writes `docs/04-calcs/results.csv`.

The design case is a 1.0 m caisson ring with a 75 mm wall inside a 1.3 m lining, the digging face 10 m below ground (R4) and 3 m of water standing over it at the end of the work (R1). The HatchSide tripod stands over the well with SinkGrab's line in place of its winch.

■ Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Sand	Saturated fine to medium sand 2.0 kg/L (1.0 kg/L submerged); 3 L of water still in the grab out of the water	Handbook ranges
Bite	Shells fill to 75 % in loose saturated sand (range 50 to 90 %)	Estimate; settled only in test-pit trials
Hoisting	1.1 dynamic allowance; 50 W per person sustained at a crank; a 400 N short heave	Ergonomic ranges
Pins	S235 bar or mild steel wire, ultimate shear strength 240 MPa; crank pin scatter ± 20 %; link pins break-tested from one coil, ± 10 %	0.6 times 400 MPa tensile
Tripod	HatchSide: 150 kg safe working load for material handling; proof 1.5 times	HTS-CAL-001
Rope	8 mm polyester double braid, 12 kN minimum breaking strength; eye splice keeps 90 %	Typical catalogue figures
Drive	Chain 95 %, each bearing 98 %	Typical figures
Brake	Lining on steel, friction 0.35; 270 degree wrap	Typical figures
Rings	Precast concrete 2,400 kg/m ³ ; string of 8 rings, 6 under water; skin friction 1 to 3 kPa in loose saturated sand disturbed by the undercut	Estimate; no published figure for small well rings
Scraper	150 N per toe cutting a 25 mm bite of loose saturated sand	Estimate
Cycle	Lowering on the brake at 0.5 m/s; cranks off and on 20 s; opening 10 s; closing 15 s; doors and dumping 45 s	Estimates
Tape	One dip-tape reading at depth ± 3 mm	Estimate

■ A. Grab capacity and mass (R3, R9)

The closed shells, 230 mm in radius and 340 mm wide, hold 28.3 L; at a 75 % fill a bite is 21.2 L [A1]. The grab weighs 47.9 kg: the shells 11.6 and 11.8 kg, the head 8.1 kg with 5.3 kg of ballast plates, the tie rods 3.8 kg [A2]. Full and out of the water it is 93.3 kg (915 N) [A3]; under water 63.0 kg (618 N) [A4]. With the 1.1 allowance the working line pull is 1,006 N [A5].

■ B. Closing force (R3)

Closing moves the crosshead 142 mm toward the head, which takes 284 mm of line through the 2:1 tackle, while the lips travel 401 mm in all; opened, the lips are 352 mm apart [B1]. The mean force at the lips is therefore 0.71 of the line pull. The line can pull no more than the grab's submerged weight, 618 N, before the grab lifts off the bottom, so the lips close with about 437 N, 643 N per metre of lip [B2]. A 3:1 tackle would raise the factor to 1.06 and the lip force to about 656 N; the head weighs 13.3 kg with its plates, which is what opens the shells [B3]. If the fill falls to 50 % in denser sand, a bite is 14.1 L [B4]. **R3 is at risk:** it is met at the assumed fill and missed at 50 %.

■ C. Overload limits (R7)

A 3 mm S235 pin across the 25 mm crank shaft releases at 42.4 N m [C1]. Through the 4:1 chain that is a line pull of 1,756 N on the innermost layer of rope, 1,610 N on the second and 1,486 N on the third; with ± 20 % scatter the pin releases between 1,189 and 2,107 N [C2]. The HatchSide tripod's safe working load is 1,472 N and its proof load 2,207 N, so the crank pin never lets the line pass the proof load [C3]. Without the pin, two people heaving 400 N each could put 8,280 N into a snagged line [C6].

The shear link at the tackle's dead end is the precise limit. A pin of about 1.88 mm in double shear releases at 1,335 N nominal; break-tested pins from one coil, ± 10 %, release between 1,202 and 1,469 N, inside the tripod's 1,472 N rating [C4]. Its lowest release is 1.19 times the working pull [C5]. When it releases, the line runs free through the crosshead sheave and the grab opens on the bottom; it is recovered on the recovery line.

■ D. Crank force and speed (R5)

At the rated grab load on the third layer, one person pushes 104 N on a 250 mm crank and two people 52 N each [D1]. **R5 is met on paper.** Two people at 50 W each hoist the full grab at 6.0 m/min, cranking at about 40 rpm [D2].

■ E. Drum and rope (R7)

The drum holds 23 turns a layer: 12.7, 13.9 and 15.1 m in three layers, 41.7 m, against the 38.7 m a 30 m well needs [E1]. The drum stands 3.0 m from the lead sheave for a fleet angle of 1.91 degrees [E2]. The 8 mm line keeps 10.8 kN through its splice: a factor of 10.7 on the working pull and 5.1 on the crank pin's highest release [E3]. The sheaves are 16.2 rope diameters at the pitch line and the drum 22.0 [E4].

■ F. Brake and pawl

The band wraps 270 degrees, so the tight side carries 5.20 times the slack side. A 6 kg weight on a 10.2:1 lever gives 598 N on the slack side and 314 N m of holding torque [F1]. The most the line can put on the drum is 186 N m, at the crank pin's limit on the innermost layer: a factor of 1.69, and 3.0 at the working pull [F2]. The ratchet tooth carries 2,211 N at the limit, 17 MPa on its face [F3].

■ G. Tripod, cradle, drawbar and capstan stability (R7)

At the limit the lead sheave is pulled 833 N outward and 1,678 N up, and the drawbar pushes the cradle 2,107 N inward. The foot therefore sees the rope along leg A, which is the load case HatchSide was designed for with its own winch [G1]. The 2,470 mm drawbar has an Euler load of 39.4 kN, 19 times the limit [G2]. The capstan weighs 68.9 kg; because the rope runs 80 mm above the drawbar, it is pitched forward by 169 N m at the limit, against 250 N m from its weight; the stakes hold it from skating [G3]. The line's rated load is 122 kg, the link's lowest release. Proof at twice the working load is 187 kg, inside HatchSide's 225 kg proof load [G4]. **R7 is met on paper.**

■ H. Scraper (R2, R8)

Two toes cutting 150 N each at 590 mm radius need 177 N m, 136 N for each of two people on the T-bar [H1]. The 42.4 x 2.6 mm pole carries that at 28 MPa and winds up 9 degrees over 10 m [H2]. The scraper head weighs 24.6 kg, each pole section 7.1 kg and the T-bar 5.6 kg; hung at 10 m it is 66 kg [H3]. At 25 m, with 12 sections, it is 115 kg, 1,245 N with the allowance, under both the crank pin's lowest release on the innermost layer (1,405 N) and the

tripod's rating, so 25 m is the scraper's working limit [H4]. The toes reach 590 mm, 15 mm beyond a 1.0 m ring's outer face; lifted 300 mm on the sleeve they fold to 364 mm, inside the 400 mm bore of the smallest ring [H5]. **R2 is met by design**, with a longer pair of arms for 1.2 to 1.3 m rings.

■ I. Sinking and tilt (R1, R6)

A 1.0 m ring with a 75 mm wall, 0.5 m high, weighs 304 kg. A string of 8 rings with 6 under water weighs 16.4 kN effective, and eight saddles add 1.61 kN, 10 % [I1]. After 3 m of sinking the rings have 10.8 m² of outer face in the soil, which at 1 to 3 kPa of skin friction holds 10.8 to 32.5 kN [I2]. The string keeps sinking to 3 m only while skin friction stays below 1.66 kPa [I3]. **R1 is at risk**: the outcome depends on a soil property that only a trial can measure. Removing the soil for 3 m of sinking, out to the toe radius, is about 2.96 m³ [I4].

Two dip-tape readings 1,075 mm apart resolve the tilt to ± 4.2 mm, 1 in 253; 1 in 80 is 13.4 mm, so the readings can see it clearly [I5]. Whether scraping the high side and moving saddles corrects it is a trial question. **R6 cannot be verified at TRL 3**.

■ J. Cycle time and output (R4)

A cycle at 10 m takes 22 s to lower, 20 s to take the cranks off and put them back, 10 s to open, 15 s to close, 111 s to hoist and 45 s for the doors and dumping: 3.7 min [J1]. With three people at the cranks the hoist takes 74 s and the cycle 3.1 min [J2]. The output is 342 L an hour of sand in place, so the 3 m of sinking takes about 8.6 h of grabbing [J3]. **R4 is not met on paper**: the hoist is limited by the power two people can give, not by the gearing.

■ K. Masses and cost (R9, R11)

The heaviest pieces are the scraper head at 24.6 kg, the capstan frame at 22.9 kg, a ballast saddle at 20.6 kg, the foot cradle at 19.4 kg and the winding drum at 18.3 kg [K1]. The assembled grab is 47.9 kg, carried by two people; the longest pieces are the pole sections at 2.0 m [K2]. **R9 is met on paper**. Value-engineering target: USD 4,000. Estimated cost of the constructable design: USD 2,506.00 (USD 1,494.00 under the target) [K3]. The kit weighs about 452 kg without ropes, tubs and the tripod [K4].

■ M. Results against every requirement

Table 2. Requirement status from this note [M].

ID	Requirement	Value	Target	Status
R4	Cycle time at 10 m, two operators	3.7 min (3.1 min with three at the cranks)	3 min or less	Not met on paper
R1	Water depth reached without de-watering	Sinks while skin friction stays below 1.66 kPa; estimate 1 to 3 kPa	At least 3 m	At risk
R3	Grab load per bite	21.2 L at 75 % fill; 14.1 L at 50 %	At least 20 L	At risk
R6	Tilt	Readings resolve 1 in 253; correction by scraping the high side and saddles	1 in 80 or better	Not verifiable at TRL 3
R5	Crank force	52 N each with two people (104 N for one)	150 N or less	Met on paper
R7	Proof load of lifting parts	Proof 187 kg inside HatchSide's 225 kg; line factor 10.7	2 times working load	Met on paper
R9	Portability	Heaviest piece 24.6 kg; kit about 452 kg; longest 2.0 m	25 kg or less; small pickup	Met on paper
R11	Prototype cost	USD 2,506	Value-engineering target USD 4,000	Met on paper, within the target
R2	Fits rings of 0.8 to 1.3 m	Grab 468 mm across; scraper folds to 364 mm radius; adjustable toes and skids, longer arms	0.8 to 1.3 m	Met by design
R8	No person in the well	Every task from the surface; doors cover the shaft	100 %	Met by design
R10	Local build	Steel section and plate, stick welding, drilling; shell skins rolled	Stick welder and hand tools	Met by design

Counts: 1 not met, 2 at risk, 1 not verifiable at TRL 3, 4 met on paper, 3 met by design. The three that are not met or at risk are posed to Amish as decisions in SKG-DEC-001.

■ Checks against the TRL 1 concept

Concept (SKG-PRC-001 v0.1)	This note	Action
Rope-closed grab with a closing line and a holding line	Two lines need two hoists and two people in step	One line; the head's weight opens the shells (SKG-DDR-001)
Hand capstan on one tripod leg	A 69 kg capstan cannot hang on an aluminium leg	Ground capstan with a drawbar to a cradle under the leg A foot (SKG-DDR-002)
HatchSide tripod "can be rated" for the loads	The tripod is rated 150 kg; two people at a crank can reach 8.3 kN	Shear link and crank shear pin (R7)
Scraper on a pole or guided frame	A pole is the only way to turn a tool at depth	Centred pole with a centralizer, arms opened by weight
Spoil chute and tipping hook	A chute over the mouth must be moved every cycle and cannot cover the shaft	Folding doors and a spoil tub